

## Trainings ▾

### General Information

Refer to Procedure 375-004 (/qs3/pubsys2/xml/en/procedures/00/00-375-004.html) in Section 1 for Biodiesel Fuel.

Refer to Procedure 375-002 (/qs3/pubsys2/xml/en/procedures/00/00-375-002.html) in Section 1 for Paraffinic Fuel.

Refer to Procedure 375-003 (/qs3/pubsys2/xml/en/procedures/00/00-375-003.html) in Section 1 for Marine Distillate Oils.

Refer to Procedure 375-005 (/qs3/pubsys2/xml/en/procedures/00/00-375-005.html) in Section 1 for Common Issues and Resolutions.

#### Cummins Inc. Required Diesel Fuel Specifications

Fuels meeting national and international specifications can be used if they conform to the specifications listed in Table 1. Cummins® engines will operate satisfactorily when using fuels meeting all the properties listed in Table 1. Fuels produced to the United States' specification ASTM D975, China's fuel specifications GB19147 and GB252, the European Union specification EN590, Indian specification IS 1460, Australia's Fuel Standard Determination 2019, Japan's JIS K 2204, Canada's CAN/CGSB-3.517 Type A and B ULS, Brazil's ANP Resolution 50/2013 S10 specification, and Russia's GOST R32511-2013 S10 specification meet Table 1 requirements and are approved for use. Where discrepancies arise, Table 1 shall be considered the minimum requirement.

**Table 1, Cummins Required Fuel Properties**

| Property                  | Units                      | Limit | Method     | Value          |
|---------------------------|----------------------------|-------|------------|----------------|
| Flash Point <sup>1</sup>  | °C [°F]                    | Min.  | ASTM D93   | 38 [ 100.4 ]   |
|                           |                            |       |            | 52 [ 125.6 ]   |
| Density                   | g/cc [lb/in <sup>3</sup> ] | Min.  | ASTM D4052 | 0.816 [ .029 ] |
|                           |                            | Max.  |            | 0.876 [ .032 ] |
| Distillation <sup>2</sup> | °C10 % [°F10%],<br>% vol   | Min.  | ASTM D86   | 150 [ 302 ]    |
|                           | °C90 % [°F90%],<br>% vol   | Min.  |            | 282 [ 539.6 ]  |
|                           |                            | Max.  |            | 338 [ 640.4 ]  |
| Cleanliness <sup>3</sup>  | -                          | -     | ISO 4406   | Code 18/16/13  |
| Ash                       | % mass                     | Max.  | ASTM D482  | 0.01           |

**Table 1, Cummins Required Fuel Properties**

| Property                         | Units         | Limit | Method                   | Value                                                                      |
|----------------------------------|---------------|-------|--------------------------|----------------------------------------------------------------------------|
| Sulfur                           | ppm           | Max.  | ASTM D5453<br>ASTM D2622 | See corresponding Owners Manual. Reference Procedure 018-002 in Section V. |
| Copper Strip                     | rating        | Max.  | ASTM D130                | No. 3                                                                      |
| Cetane Number <sup>4</sup>       | -             | Min.  | ASTM D613                | 42                                                                         |
| Kinematic Viscosity <sup>5</sup> | cSt           | Min.  | ASTM D445                | 1.3                                                                        |
|                                  |               | Max.  |                          | 4.1                                                                        |
| Water                            | ppm           | Max.  | ASTM D6304<br>ISO 12937  | 200                                                                        |
| FAME content                     | % vol         | Max.  | EN 14078                 | 20                                                                         |
| Cloud Point                      | °C [°F]       | Max.  | ASTM D2500               | Should meet lowest expected ambient temperature                            |
| Lubricity (HFRR)                 | micron [inch] | Max.  | ASTM D6079               | 520 [ 0.0205 ]                                                             |
| Sodium Content                   | ppm           | Max.  | ASTM D7111               | 0.5                                                                        |
| Chlorine Content                 | ppm           | Max.  | ASTM D4929               | 10                                                                         |

1. The minimum Flash Point temperature for #1 Diesel is 38 °C [ 100.4 °F ]. The minimum Flash Point temperature for #2 Diesel is 52 °C [ 125.6 °F ].
2. The minimum temperature for 90% of fuel to be recovered for #2 diesel is 282 °C [ 539.6 °F ]. The maximum temperature for 90% of fuel to be recovered for #2 diesel is 338 °C [ 640.4 °F ]. The maximum temperature for 90% of fuel to be recovered for #1 diesel is 288 °C [ 550.4 °F ].
3. Fuel that is being dispensed into equipment fuel tank is strongly recommended to meet cleanliness specifications listed in ISO 4406. For requirements specific to Standby Datacenter applications, see Fuel Cleanliness section.
4. For ambient temperatures above 0 °C [ 32 °F ] a minimum cetane number of 42 is required. For ambient temperatures below 0 °C [ 32 °F ] a minimum cetane number of 45 is required.
5. Kinematic viscosity for #1 Diesel should be between 1.3 and 2.4 cSt. Kinematic viscosity for #2 Diesel should be between 1.9 and 4.1 cSt.

### Diesel Fuel Properties

#### Viscosity:

- General Description - Proper viscosity provides adequate pumping and lubricating characteristics to fuel system components, and is also responsible for controlling internal

leakage in fuel pumps and injectors.

- Test Method - ASTM D445, ISO 3104

Cetane Number:

- General Description - Cetane number is a measurement of the combustion quality of diesel fuel during compression ignition. In cold weather or in service with prolonged low loads, a higher cetane number is desirable.
- Test Method - ASTM D613, ISO 5165
- Fuel with a cetane number greater than 55 can cause increased torque peak smoke. Reference American Society for Testing and Materials (ASTM) D613, ISO 5165.

Cetane Index:

- General Description - Cetane index is often used as a substitute for the cetane number of diesel fuel.
- Cetane index is calculated based on the fuel density and distillation range (ASTM D86).
- Test method - There are two methods used, ASTM D976 and D4737. The D976 method is often referred to as the two-variable equation, and is obsolete. This method should no longer be used for cetane number estimation, except as required by the United States Environmental Protection Agency (EPA) as an alternative method for satisfying an aromaticity requirement. D4737 is the present method and is often referred to as the four-variable equation. D4737 is the same method as ISO 4264.

**Note : Cetane index is not an appropriate approximation for ignition quality of fuels containing a cetane improver additive.**

Sulfur Content:

- General Description - Diesel fuels contain varying amounts of various sulfur compounds. Fuel sulfur contributes to acid formation, exhaust particulates, fuel system corrosion, and reduced oil drain intervals. Reduced sulfur is required to avoid poisoning aftertreatment devices. Higher sulfur fuel also needs higher total base number (TBN) lubricants to compensate for acid corrosion.
- Test Method - ASTM D2622, ISO 4260

**Note : Diesel fuel sulfur levels that exceed the recommendation will shorten the life of certain aftertreatment components, including but not limited to the aftertreatment diesel oxidation catalyst (DOC), aftertreatment diesel particulate filter, and the aftertreatment selective catalyst reduction (SCR). Catalyst malfunctions caused by the use of diesel fuels which exceed the recommended sulfur levels are not warrantable. General guideline is no more than 15 ppm sulfur diesel fuel if aftertreatment DOC or aftertreatment DPF are present, no more than 50 ppm if aftertreatment SCR is present and no more than 5000 ppm for engines without emissions equipment. See corresponding Owners Manual for specific engine requirements.**

Active Sulfur:

- General Description - Some sulfur compounds in fuel are actively corrosive.
- Test Method - ASTM D130, ISO 2160

Water and Sediment:

- General Description - The amount of water and solid debris in the fuel is generally classified as water and sediment. A practice of filtering the fuel while being put into the fuel tank is recommended. More water vapor condenses in partially filled tanks due to tank breathing caused by temperature changes. Filter elements, fuel screens in the fuel pump, and fuel inlet connections on injectors **must** be cleaned or changed whenever contaminated. These screens and filters, in performing their intended function, become clogged when using a poor or dirty fuel, and will need to be changed more often.
- Test Method - ASTM D1796

#### Carbon Residue:

- General Description - The tendency of a diesel fuel to form carbon deposits in an engine can be estimated by determining the Ramsbottom or Conradson carbon residue of the fuel after 90 percent of the fuel has been evaporated.
- Test Method - ASTM D524, ASTM D189, ISO 10370

#### Density:

- General Description - Density is an indication of the energy content of the fuel. Higher density indicates more thermal energy and better fuel economy.
- Test Method - ASTM D287, D4052, ISO 3675

#### Cloud Point:

- General Description - The cloud point of the fuel is the temperature at which crystals of paraffin wax first appear. Crystals can be detected by the cloudiness of the fuel.
- Test Method - ASTM D2500, ISO 3015

#### Cold Filter Plugging Point:

- General Description - The cold filter plugging point of the fuel is the lowest temperature at which fuel can still flow through a 45 micron wire mesh. This test method can be directly related to a fuel's tendency to plug fuel filters at reduced temperatures, due to the formation of paraffin wax crystals.
- Test Method - ASTM D6371

#### Ash:

- General Description - The small amount of noncombustible metallic material commonly found in almost all petroleum products is called ash.
- Test Method - ASTM D482, ISO 6245

#### Distillation:

General Description - The distillation characteristics of a particular fuel gives important information about the composition and behavior of the fuel during storage and use by measuring the rate of evaporation.

Test Method - ASTM D86, ISO 3405

#### Lubricity:

- General Description - Lubricity is the ability of a liquid to provide hydrodynamic and boundary lubrication to prevent wear between moving parts. Fuel with lower sulfur and viscosity tends

to have lower lubricity. Lubricity can be measured by either one of two procedures.

- Test Method: ASTM D6078, Scuffing Load Ball On Cylinder Evaluator (SLBOCLE), or ASTM D6079, ISO 12156, High Frequency Reciprocating Rig (HFRR)

Sodium Content:

- Sodium based contaminants are known contributors to deposit formation as a result of adverse reactions with fuel additives. These deposits hinder proper function of fuel injection equipment
- Test Method: ASTM D7111

### Ultra-Low Sulfur Diesel

Ultra-low sulfur diesel (ULSD) fuel is defined as diesel fuel **not** exceeding 0.0015 mass percent (15 ppm) sulfur content. The following Cummins® engines are **required** to operate on ULSD.

- Engines operating with aftertreatment systems including: diesel particulate filters, diesel oxidation catalysts, and selective catalytic reduction. Reference the engine specific Owner's Manual, Procedure 018-002 (Fuel Recommendations and Specifications) in Section V, for fuel sulfur requirements.
- Engines operating where regional, national, or international regulations require the use of ULSD in diesel engines.

In general ULSD regulations make the fuel compatible with diesel engines with higher tolerance for sulfur content. The low sulfur content does slightly change some of the fuel properties. When transitioning from low sulfur diesel (LSD), which can contain up to 500 ppm sulfur, to ULSD the following precautions are beneficial:

- Properly label all fuel tanks and delivery pumps.
- Be sure that the fuel lubricity meets the specifications outlined in Table 1: Cummins Inc. Required Diesel Fuel Specifications in this bulletin.
- ULSD has lower lubricity than LSD, so lubricity additives need to be added by the fuel supplier to prevent fuel system damage. More information on fuel additives can be found in the "Additives" section of this service bulletin.
- Be sure that fuel tanks are completely empty before transitioning from LSD to ULSD, and consider tank cleaning.
- It takes **only** a small amount of LSD blended with ULSD to bring the fuel sulfur content above 15 ppm.
- Consider using a stability additive for fuel in bulk storage.
- ULSD is more prone to oxidation than LSD. Consult your fuel supplier to determine if an additive is needed to maintain fuel quality in storage tanks.
- Closely monitor the fuel system for leaks, especially when first transitioning to ULSD, and correct them immediately.
- ULSD reacts differently than LSD with certain seal and gasket compounds commonly found in fuel systems, which means that leaks are more likely to occur, especially in older engines which were designed to run on LSD.

Fuel Dyes

It is common in many regions for fuel to be dyed, most often to identify its tax status. The fuel is most commonly dyed red, but other colors are used as well. Diesel fuel dye does **not** affect the chemical composition of the fuel and is sometimes used to distinguish it from fuels of differing grades, specifications, sulfur content, etc. Consult the fuel supplier to understand the significance of any dyes used in the fuel purchased.

Dyed diesel fuel can be used in Cummins® engines, provided the fuel meets all applicable regional, national, and international regulations for the engine application. The fuel **must** also meet or exceed the specifications outlined in Table 1: Cummins Inc. Required Diesel Fuel Specifications in this bulletin.

## Fuel Cleanliness

This section explains the importance of fuel cleanliness to the successful operation of Cummins® Engines.

Modern fuel systems have been developed to reduce emissions and fuel consumption, and improve engine performance. These high-pressure systems operate at pressures approaching 2100 bar [ 30,500 psi ] and with component match clearances typically from 2 to 5 microns for injectors. At these pressures, very small, hard particles are potential sources of fuel system malfunction.

Excessive contamination of diesel fuel can cause premature clogging of diesel fuel filters and/or premature wear of critical fuel injection system parts. Depending on the size and nature of the particles, this can lead to:

- Reduced component life.
- Component malfunction.
- Fuel system and/or engine failure.

Determining fuel cleanliness requires measuring both the size and number of particles per size class in the fuel, i.e. the particle size distribution. The International Standards Organization (ISO) has developed a protocol for expressing the level of contamination by coding the size distribution called ISO 4406.

ISO 4406 cleanliness codes are expressed as a series of three numbers (x/x/x), which correspond respectively to the number of particles greater than 4, 6, and 14 microns.

- 21 - Up to 20,000 particles larger than 4µm (per mL of fuel)
- 18 - Up to 2,500 particles larger than 4µm (per mL of fuel)
- 16 - Up to 640 particles larger than 6µm (per mL of fuel)
- 13 - Up to 80 particles larger than 14µm (per mL of fuel)

Engine builders and fuel injection equipment manufacturers have found that the particles of size approaching the 4 and 6 micron ISO checkpoints are particularly critical to the durability of the fuel injection system.

Cummins Inc. recommends that if the fuel does **not** meet the ISO cleanliness code of 18/16/13 when supplied to the engine, additional filtration be applied before the fuel is delivered to the equipment's fuel tank. A Cummins® Distributor or Cummins Filtration™ representative can supply

hardware and additional filtration guidance and can recommend countermeasures such as improved fuel quality from the fuel supplier, and/or better fuel handling, storage, dispensing, and fuel tank cleaning techniques.<sup>1</sup>

1. For EPA Tier 2 Standby Datacenter applications, QSK60, QSK78 and QSK95 engines have been approved to use ISO 4406 21/x/x fuel. To be applicable, filtration requirements listed in engine O&M manuals **must** be met or exceeded.

## **Fuel Sampling**

Cummins Inc. recommends collection and analysis of diesel fuel at periodic intervals to make sure bulk fuel supply continues to meet all of the requirements in Table 1 and provides adequate performance and protection of the engine and components. Cummins Inc. sampling frequency recommendations are applicable to all engines, all markets, and all fuel types; however the practicality of maintaining as delivered fuel quality is most applicable to customers utilizing bulk fuel delivery and storage systems. The optimal frequency and level of analysis recommended will differ based on equipment type, user application, duty cycle, region, and environmental considerations. Given the large number of variables to consider, Cummins Inc. recommends all customers use the below schedule as best practice, but work with the fuel supplier to custom tailor a fuel sampling and analysis program best suited for the situation.

## **Sampling Frequency and Methods**

Several critical environmental, chemical, and handling events should be used to initiate fuel sampling and analysis. At a minimum, Cummins Inc. recommends fuel sampling and analysis be completed quarterly and/or after a full bulk inventory turn (cycle), whichever comes first.

Additional triggers to consider include:

- New/adjusted fuel blends and change overs (summer vs. winter fuel)
- Addition/changes to additive packages (injector deposit cleaner, cetane improvers, lubricity improvers, etc.)
- Change of fuel supplier
- New installations or customer operations
- Change over from LSD to ULSD
- Blending/concentration change of biodiesel
- Significant weather changes (rapid temperature swings, dust storms, heavy rain/snow, etc.)

Fuel sampling techniques will depend on the location within the fuel supply chain to be sampled. Cummins Inc. recommends fuel samples be taken directly from vehicle/equipment at the inlet to suction side fuel filter. Cummins Inc. recommends use of Cummins Filtration<sup>TM</sup> MONITOR Fuel Analysis Test packages. Contact the local Cummins Distributor, or Cummins Filtration representative for more information.

## **Analysis Recommendations**

Cummins Inc. recommends fuel analysis be completed by an ISO/IEC 17025 certified laboratory, with experience in diesel fuel analysis and capability to perform tests defined in Table 1. The quantity and type of test to be performed will determine the cost of the associated sampling. It is the customer responsibility to understand and maintain the quality of the fuel used in their engine. A basic list of fuel sample analysis tests follows in Table 2 below.

| Table 2, Recommended Fuel Sample Test Properties                                                                                                                                                                                                                                                                                                                                                                                          |                                        |                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Property/Item                                                                                                                                                                                                                                                                                                                                                                                                                             | Test Priority (A or B) & Frequency     | Purpose/Importance                                                                                                 |
| Particle Count (ISO 4406)                                                                                                                                                                                                                                                                                                                                                                                                                 | A – Every Sample                       | Prevent rapid erosion of fuel injection equipment – poor performance and component failure.                        |
| Water Content (ASTM D6304)                                                                                                                                                                                                                                                                                                                                                                                                                | A – Every Sample                       | Corrosion prevention, rapid fuel injection equipment wear-out mechanisms. Prevention of deposit formation (IDID's) |
| ICP Metals                                                                                                                                                                                                                                                                                                                                                                                                                                | A – Every Sample                       | Prevention of deposit formation (IDID's), indicates presence of lube oil in fuel.                                  |
| Cloud Point/Pour Point                                                                                                                                                                                                                                                                                                                                                                                                                    | A – Annually, or prior to cold season. | Prevent fuel filter and system plugging due to waxing/gelling                                                      |
| Sulfur Content                                                                                                                                                                                                                                                                                                                                                                                                                            | A – Every Sample                       | Prevent premature fuel injection equipment wear out. Fuel and lube oil system corrosion prevention.                |
| Simulated Distillation                                                                                                                                                                                                                                                                                                                                                                                                                    | A – Every Sample                       | Prevent premature fuel engine component wear out due to cavitation or lube oil dilution.                           |
| Thermal Stability                                                                                                                                                                                                                                                                                                                                                                                                                         | A – Every Sample                       | Prevent premature fuel injection equipment wear out. Corrosion prevention.                                         |
| Biological Contamination                                                                                                                                                                                                                                                                                                                                                                                                                  | B - Quarterly                          | Corrosion prevention, premature fuel filter plugging.                                                              |
| HFRR (ASTM D 6079                                                                                                                                                                                                                                                                                                                                                                                                                         | B – Quarterly                          | Prevent premature fuel injection equipment wear out.                                                               |
| ISO 12156/1)                                                                                                                                                                                                                                                                                                                                                                                                                              |                                        |                                                                                                                    |
| Tank Vent Filtration                                                                                                                                                                                                                                                                                                                                                                                                                      |                                        |                                                                                                                    |
| Particles in the 4 to 6 micron size range require laboratory equipment to identify, yet can do significant damage to high pressure fuel systems when the cleanliness of the fuel in the tank exceeds the ISO 4406 code 18/16/13 maximum. Cummins Inc. recommends that all fuel tanks be fitted with a tank vent filter (of at least 98.7 percent efficiency at 10 micron) to prevent dirt from entering the tank as the fuel level drops. |                                        |                                                                                                                    |
| Stand-by and Emergency Power Generation                                                                                                                                                                                                                                                                                                                                                                                                   |                                        |                                                                                                                    |



Engines intended to supply stand-by or emergency power present unique situations for fuel quality and cleanliness. These engines are **not** used frequently, and therefore could possibly require special considerations for fuel handling and storage.

The engine manual discusses the specific procedures for maintaining the engine in a state of readiness. This section is concerned with the fuel supply.

Fuel tanks **must** be inspected and maintained to avoid contamination of the fuel by either water or dirt. Consult with your fuel supplier for qualified persons or laboratories in your area to help with monitoring of the fuel supply. Samples can be taken from the top, middle, and bottom of the tank every 6 months and checked for cleanliness and biological contamination, as well as to make sure the fuel still meets the specifications in Table 1: Cummins Inc. Required Diesel Fuel Specifications in this service bulletin.

Long term storage (in excess of 6 months) is **not** recommended unless the fuel has been stabilized by the fuel supplier and there is a monitoring program in place. Periodic testing of the engine is recommended to be performed frequently enough and long enough to make sure that the fuel supply is replenished and stays fresh.

#### Duplex Fuel Filtration Systems

Stand-by and emergency generators can be called upon to run for hundreds of hours in case of emergency. Such critical operations could possibly consider installation of a duplex fuel filtration system. These systems allow rapid switching to fresh fuel filters. It is recommended that such service occur while the engine is shut down briefly. A Cummins® or Cummins Filtration™ Distributor can advise on the proper installation for a particular engine and location.

#### Fuel Tank Care and Maintenance

Tank cleaning is a major operation which requires complete draining of the tank, and should **only** be done by professionals. It is therefore carried out infrequently, normally on the schedule of several years, coinciding with (statutory) inspection and maintenance requirements. Good housekeeping can help extend periods between tank cleaning.

Water bottom measurements can be made on an appropriate time interval (via automatic gauging or regular tank dipping with water finding paste) and water can be removed when necessary. This is important since any water and sediment can be stirred up when the tank is filled. Cummins Inc. recommends waiting a minimum of one hour per foot of fuel depth before dispensing fuel after a delivery. If water and sediment are observed, additional settling time is one way of bringing the fuel back into specification.

It is virtually impossible to stop water from entering the supply chain; therefore, good housekeeping is essential. Hardware, tanks, and pumping systems should be routinely inspected and maintained. Fuel should be checked periodically for contamination by water to make sure that there is no free water present in the fuel entering the engine, and dissolved (emulsified) water does **not** exceed 200 ppm.

Cummins Inc. recommends that if the fuel does **not** meet the ISO 4406 cleanliness code of 18/16/13 in bulk storage, additional filtration be applied before the fuel is delivered to the engine. A Cummins® or Cummins Filtration™ Distributor can supply hardware and additional filtration guidance.

Contingency Diesel Fuel Specifications

This section presents the specifications for fuels which are **only** to be used when fuel meeting the required specifications is **not** available. In the case that fuels meeting the specifications in Table 1: Cummins Inc. Required Diesel Fuel Specifications are **not** available, Cummins Inc. has prepared contingency specifications to aid the user in choosing the most acceptable contingency fuel.

**⚠ CAUTION ⚠**

**Fuels outside the recommended fuel specifications, but within the contingency specifications, are only meant to be used for short periods of time when no other fuels are available. Use of contingency fuels can have an adverse effect on engine performance and durability. Cummins Inc. assumes no warranty responsibility for repairs or increased costs of operation resulting from the use of fuels that do not conform to the specifications listed in Table 1.**

Guidelines for the Use of Contingency Fuels

- A calibration change of the fuel pump or injectors is **not** recommended when changing to a contingency fuel that meets all the specifications shown in Contingency Diesel Fuel Specifications, although changing to a contingency fuel can cause a slight power loss and can result in higher than normal wear of certain components. See the sections in this bulletin on Power Loss and Component Wear and Durability for additional information.
- Combustion performance may be affected when using fuels meeting contingency fuel specifications. It is the responsibility of the user to make certain that the use of contingency fuels and the subsequent change in performance does **not** exceed legal limits.
- Most jet fuel lubricities are too low to provide the necessary lubrication for the fuel system components. If (based on the fuel supplier's specifications) a fuel does **not** have the minimum lubricity listed for contingency fuels in the Contingency Diesel Fuel Specifications, a fuel additive **must** be added to the fuel to increase the lubricity and specially enhanced fuel system components **must** be used. Reference the section in this bulletin on fuel additives. Consult Cummins Inc. for available hardware options.

**⚠ CAUTION ⚠**

**Some contingency fuels, such as jet fuels and kerosene, are much more flammable than normal diesel fuel. Use extreme care to keep cigarettes, flames, pilot lights, sparks, arcing equipment and switches, and other sources of ignition away and out of areas sharing ventilation.**

Additional maintenance can be required when using contingency fuels. Those using contingency fuels **must** consult with their fuel supplier to determine any problems which can result from using fuels meeting the Contingency Diesel Fuel Specifications. If there is still a question, data on the fuel's physical properties **must** be submitted to Cummins Inc. Service Engineering Department for review before use in Cummins® engines.

**Table 3, Contingency Diesel Fuel Specifications <sup>1</sup>.**

|           |                                           |
|-----------|-------------------------------------------|
| Viscosity | 1.3 to 13.1 centistokes at 40°C [ 104°F ] |
|-----------|-------------------------------------------|

**Table 3, Contingency Diesel Fuel Specifications <sup>1</sup>.**

|                             |                                                                                                                                                                                              |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cetane Number               | 35 minimum above 0°C [ 32°F ]; 40 minimum below 0°C [ 32°F ]                                                                                                                                 |
| Sulfur Content              | Less than 2.0 mass-percent (20,000 ppm). Catalyst equipped engines will <b>not</b> be able to use high sulfur fuel even for a short period of time without permanent damage to the catalyst. |
| Active Sulfur               | Copper Strip Corrosion <b>not</b> to exceed Number 2                                                                                                                                         |
| Water and Sediment          | <b>Not</b> to exceed 0.5 volume-percent                                                                                                                                                      |
| Carbon Residue              | <b>Not</b> to exceed 5.0 mass-percent on 10 volume-percent residuum                                                                                                                          |
| Density                     | 0.750 to 0.965 g/cc at 15°C [ 59°F ]                                                                                                                                                         |
| Cloud Point                 | 6°C or 11°F below lowest ambient temperature at which the fuel is expected to operate                                                                                                        |
| Ash                         | <b>Not</b> to exceed 0.05 mass-percent                                                                                                                                                       |
| Distillation                | 90 volume-percent at 395°C [ 743°F ]                                                                                                                                                         |
| Lubricity (HFRR or SLBOCLE) | HFRR: Maximum of 0.6 mm Wear Scar Diameter (WSD) at 60°C [ 140°F ].<br>SLBOCLE: Minimum of 2300 grams <sup>2</sup> .                                                                         |
| Vanadium                    | 5 ppm, maximum                                                                                                                                                                               |
| Aluminum                    | 1 ppm, maximum                                                                                                                                                                               |
| Silicon                     | 1 ppm, maximum                                                                                                                                                                               |
| Sodium                      | 10 ppm, maximum                                                                                                                                                                              |

<sup>1</sup> Reference test methods in Diesel Fuel Properties.

<sup>2</sup> A lubricity additive **must** be used if the fuel does **not** meet the minimum lubricity specification.

#### Effects of Contingency Diesel Fuels on Engine Operation

##### Viscosity:

- Low viscosity causes rapid wear of fuel pumps and injectors, and increases internal leakage in fuel pumps and injectors. High viscosity causes hard starting, white smoke when cold, injector cup cracking, and injector train failures. Governor wear on rotary fuel pumps can cause loss of regulation.

##### Cetane Number:

- A cetane number below 42 can cause poor starting, excessive white smoke, and poor idling. A cetane number above 55 can increase smoke at peak torque conditions.

##### Sulfur Content:

- High sulfur content increases wear in injectors, piston rings, and bearings. Use of fuels with sulfur content above 5000 ppm requires the use of higher total base number (TBN) lubricants (TBN greater than 10) and shorter oil drain intervals.

**⚠ CAUTION ⚠**

**Catalyst failures caused by the use of fuels with higher than recommended sulfur levels are not warrantable. High sulfur fuel will also shorten the life of certain components in the exhaust system, including the oxidation catalyst.**

Active Sulfur:

- Excessive active sulfur increases the corrosive attack on the fuel pump, injectors, and other fuel system components.

Water and Sediment:

- Contaminated fuels reduce filter life, fuel system life, and cause on-road failures.

Carbon Residue:

- High carbon residue causes increased combustion chamber carbon deposits, more exhaust smoke, and higher soot contamination of the lubricating oil.

Density:

- Lighter fuels contain less thermal energy per gallon and result in somewhat lower fuel economy. A fuel with a density of 0.876 g/cc contains about 3.5 percent more energy per gallon than a fuel with a density of 0.815 g/cc.

Cloud Point:

- Operating below the cloud point temperature can cause the fuel filter to clog with wax crystals, restrict fuel flow, and cause loss of power. It is suggested that if fuels with cloud points above the expected ambient temperatures are purchased, the consumer **must** consult the fuel supplier and Cummins Inc. concerning fuel handling techniques. For more information, reference Common Issues With Winter Fuel.
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Pour Point:

- Operating near or below the pour point will cause start-up issues. It is doubtful that most fuel pumps can operate at the pour point. In fact, it is recommended that systems be operated at 5.5 to 8°C or 10 to 15°F above the pour point of a fuel.

Cold Filter Plugging Point

- Operating below the cold filter plugging point temperature will cause the fuel filter to clog with wax crystals, restrict fuel flow, and cause loss of power. It is suggested that if fuels with cold filter plugging points above the expected ambient temperatures are purchased, the consumer **must** consult the fuel supplier and Cummins Inc. concerning fuel handling techniques. For more information, reference Common Issues with Winter Fuel.

Ash:

- High ash content causes deposits of noncombustible metallic material in the combustion chamber and on the exhaust valves.

#### Distillation, Maximum:

- Fuels with high distillation temperature can leave gummy deposits in the fuel system and result in poor fuel combustion.

#### Lubricity:

- Fuels with low lubricity cause increased wear and possible seizure of fuel system components.

#### Vanadium:

- Fuels with high vanadium content can cause valve burning.

#### Aluminum:

- Fuels with high levels of aluminum can cause premature ring and liner wear, which can lead to excessive oil consumption.

#### Silicon:

- Fuels with high levels of silicon can cause premature ring and liner wear, which can lead to excessive oil consumption.

#### Sodium:

- Fuels with high levels of sodium can cause premature ring and liner wear, which can lead to excessive oil consumption. Sodium can combine with vanadium, if present, and catalyze, causing valve burning.

#### Zinc:

- Fuels with high levels of zinc can cause injector spray hole carboning. Do **not** use galvanized pipe or fittings in the fuel system plumbing. Diesel fuel will leach zinc galvanized material.

This section gives guidelines on power loss to be expected when using recommended or contingency fuels, or fuels that are above normal temperature.

**Note : The values given concerning power loss due to the use of contingency fuels are intended only to help estimate power loss. Power loss can vary greatly, depending on operating conditions, engine type, fuel system type, fuel composition, and other factors. These guidelines can not be used to precisely calculate engine power loss.**

The use of contingency fuels can cause a decrease in the power output of the engine, due to differences in fuel density and viscosity. In addition, changes in fuel temperature also affect engine power output, because temperature affects both viscosity and density.

#### Density

All engines will have a predictable variation in power output, depending on the density of the fuel used. Engines using fuels with a high density will produce more power than those using fuels with a lower density, because the thermal energy content of the fuel is higher. Since fuel is marketed by volume, lower density fuel carrying less thermal energy results in a proportional decrease in fuel economy or power output.

## Viscosity

In general, lower viscosity results in lower power, due to increased internal leakage in the fuel system. Also, lower viscosity fuels generally have lower thermal energy content. The effect viscosity has on power depends on the type of fuel system used.

## Temperature

Temperature causes changes in engine power because it affects both viscosity and density. An increase in fuel temperature will cause a decrease in viscosity, which will reduce power due to internal leakage in the fuel system, as described above. The maximum recommended fuel pump inlet temperature for Cummins® engines is 70°C [ 158°F ].

An increase in fuel temperature will also cause a decrease in fuel density (increase in API gravity), which will reduce power due to lower energy content of the fuel. On Cummins® engines using the PT™, Quantum®, or HPI® fuel systems, the power loss due to increasing temperature is less than that on engines using the in-line, distributor, or SELECT™ systems (less than 1 percent per 5.5°C [ 10°F ]), due to the inherent viscosity compensating characteristics of these systems.

## Component Wear and Durability

This section shows the effects of contingency fuels on wear and durability of fuel systems components.

The use of contingency fuels can affect the wear and durability of both fuel pump and injector components within the fuel system. Many of these fuels are low in viscosity and lubricity, as measured in the Ball On Cylinder Lubricity Evaluator (BOCLE) or the High Frequency Reciprocating Rig (HFRR) tests. Fuels with low lubricity can cause failure of fuel system components. Other factors that affect wear and durability are sulfur, water, and sediment content. High sulfur content increases wear of the fuel system components. Abnormal quantities of water and sediment in the fuel will also cause excessive wear, as well as other engine problems.

## Hot Restarts

This section shows how contingency fuels affect the ability of the engine to restart while still hot.

On Cummins® engines which use a distributor type fuel system, the use of contingency fuels can cause difficulty restarting the engine while it is still hot. In addition, if excessive wear exists in the fuel pump, the same difficulty can occur even when using fuels within the range listed in Required Diesel Fuel Specifications. The problem is caused by excessive leakage of fuel around the internal components of the fuel pump. Fuel leakage becomes excessive due to the high temperatures and low viscosity of the fuel. Excessive wear of the fuel pump components will make the problem worse. The leakage can become so great that the pump will **not** produce the fuel rate necessary to restart the engine. If this problem is encountered, it can be corrected by using fuel which meets the specifications in the Required Diesel Fuel Specifications section of this bulletin. If this does **not** correct the problem, repair or replacement of worn fuel pump components is necessary.

Alternate or contingency fuels can cause difficulty restarting a hot engine. The hot restart complaint can be caused by fuel burning prematurely during the first compression stroke. Lighter alternate or contingency fuels can enter the cylinder through an open injector caused by the

thermal expansion that occurs during the heat soak after engine shutdown. The burning fuels increase the starting cylinder pressure and increase the amount of torque needed to start the engine. Lighter alternate or contingency fuels with lower flash points increase the probability of fuel entering and burning in the cylinder. This issue can, on occasion, occur when using fuels that meet the specifications listed in Table 1: Cummins Inc. Required Diesel Fuel Specifications. Various Hot Restart kits (sometimes referred to as a Hot Start Knock Kits) have been released by Cummins Inc. to address this issue.

If this complaint is encountered, it can be corrected by using fuels which meet the requirements in Table 1 of this bulletin.

### Fuel Blending

This section presents the effects of blending fuels with used and new lube oil, other fuels, and with gasoline, gasohol, or alcohol. Biodiesel fuel blends are discussed in a separate section of this service bulletin.

There are two different types of fuel blending processes referred to in this section. The first is the blending of used engine lubricating oil to reduce fuel costs and to aid in disposing of used engine oil. This section also discusses the blending of fuel and engine oil in on-highway applications. The second is the blending of heavier fuels with lighter fuels to lower the wax content, cloud point, and pour point, and thus improve cold weather operation. In addition, the effects and hazards of mixing alcohol with diesel fuel are discussed.

### Blending Fuel and Lubricating Oil

#### **⚠ WARNING ⚠**

**Some state and federal agencies have determined that used engine oil can be carcinogenic and can cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.**

#### **⚠ CAUTION ⚠**

**Never blend more than 5 percent used lubricating oil with the fuel. Do not blend other used oils with fuel, such as transmission fluid, gear case oil, and so forth. Additional oil blending restrictions are outlined in this section.**

#### **⚠ CAUTION ⚠**

**Blending fuel with lubricating oil is not allowed for Cummins® Midrange, Heavy Duty, and High Horsepower engines equipped with exhaust aftertreatment devices. Oil blending on these engines will result in engine damage.**

The blending of used lubricating oil is prohibited in some locations. United States Federal Law (40 CFR 80.522) prohibits anyone from adding used lubricating oil, or used lubricating oil blended with diesel fuel, into the fuel system of model year 2007 or later diesel motor vehicles or model

year 2011 or later nonroad diesel engines (not including locomotive or marine diesel engines). Other local laws may apply and should be consulted prior to blending used lubricating oil.

Cummins Inc. provides the following guidelines for blending lubricating oil with fuel:

- Engines required to use ultra-low sulfur diesel fuel (15 ppm sulfur maximum) are **not** allowed to blend used lubricating oil with diesel.
- Engines equipped with an exhaust aftertreatment device, such as an oxidation catalyst, diesel particulate filter, or SCR system, are **not** allowed to blend used lubricating oil with diesel fuel.

All other Cummins® engines which do **not** fall in to the above categories may blend used lubricating oil with diesel fuel up to a maximum volume-concentration of 5 percent.

However, if the blended fuel exceeds the maximum regional or national sulfur content, the law may have been violated and penalties can be assessed. To be sure that blended fuel complies with the law, the sulfur level of the final blend should be tested and compared to local and national regulations.

#### Blending Fuel with Fuel

Cummins Inc. recommends the use of a premium diesel fuel during winter (ambient conditions at -7°C or 20°F or below) operating conditions. Blended fuel **must** meet the specifications in Table 1: Cummins Inc. Required Diesel Fuel Specifications. See the Additives section in this service bulletin.

In cold-weather operation, the most common method of preventing fuel waxing problems is to dilute heavier, higher wax content fuels such as diesel number 2 (D2) with lighter, lower wax content fuels such as diesel number 1 (D1) or jet fuel. This reduces the concentration of wax and thereby reduces both the cloud point and pour point. Blended fuels of this nature are more expensive to use both because they cost more and because they have a lower thermal energy content. A typical blended fuel contains 30 to 60 volume-percent light distillate fuel, usually yielding a 3 to 7°C or 5.4 to 12.6°F drop in cloud point, and a 5 to 11°C or 9 to 20°F drop in pour point. Lower wax content fuels **must** be added BEFORE wax forms to be effective.

#### Blending Fuel with Gasoline, Gasohol, and Alcohol

##### **WARNING**

**Do not mix gasoline, alcohol, or gasohol with diesel fuel. This mixture can cause an explosion.**

##### **WARNING**

**Under no circumstances must gasoline or alcohol be used to dilute diesel fuel. This practice creates an extreme fire hazard and under certain circumstances an explosive hazard. Gasoline dilution is not an effective way to lower cloud point (20 volume-percent gasoline only lowers cloud point 4°C or 7°F and it lowers the fuel viscosity, cetane number, and flash-point). Alcohol dilution will increase the cloud point.**



Alcohol is considered a renewable energy source. Some suppliers integrate up to 15 percent alcohol in diesel fuel to form oxy-diesel or e-diesel. While the use of special additives addresses some of the problems with alcohol blending in diesel fuel, Cummins Inc. recommends against the use of such blends due to safety reasons. This kind of fuel is considered experimental and is **not** covered by warranty. Engine damage, service issues or performance problems that occur due to the use of these products are **not** considered a defect in workmanship or material as supplied by Cummins Inc. and can **not** be compensated under the Cummins Inc. warranty.

## Fuel Additives

This section gives information on the use of fuel additives in Cummins® engines, including water emulsifiers.

Cummins Inc. neither approves nor disapproves of the use of any fuel additive, fuel extender, fuel system modification, or the use of any device **not** manufactured or sold by Cummins Inc. or its subsidiaries. Engine damage, service issues, or performance problems that occur due to the use of these products are **not** considered a defect in workmanship or material as supplied by Cummins Inc. and can **not** be compensated under the Cummins Inc. warranty. Cummins® engines are designed, developed, rated, and built to operate on commercially available diesel fuel as listed in Table 1 in this bulletin.

Generally, aftermarket additives are unnecessary. There are certain conditions where fuel additives can provide benefits. Cummins Inc. **only** endorses two additives for these situations. PowerService Diesel Kleen® +Cetane Boost® and PowerService Diesel Fuel Supplement® +Cetane Boost®.

Cummins Filtration™ offers lubricity enhancing fuel filters that can improve the lubricity of fuels that fall below the recommended lubricity specification given in Table 1: Cummins Inc. Required Diesel Fuel Specifications. The following filters are required by Cummins Inc. when the corresponding engine is operated using low lubricity fuels such as Jet A, JP5, or JP8.

**Note : Refer to Procedure 018-002 in Section V in the corresponding owners manual to ensure the fuel meets other specific engine requirements.**

| Table 4, Cummins Filtration™ Lubricity Enhancing Filters |                           |                                                                              |
|----------------------------------------------------------|---------------------------|------------------------------------------------------------------------------|
| Filter                                                   | Fuel System Compatibility | Engine Compatibility                                                         |
| FS20000                                                  | Rotary Fuel Systems       | B Series -<br>Tier II Industrial                                             |
| FS20022                                                  | Common Rail               | B, C, and L Series -<br>Tier III Industrial and Marine<br>QSF3.8 Stage III A |

Other optional lubricity enhancing spin-on dosers approved for use in Cummins® common rail engines using low-lubricity fuels include FA15700 and CD1575401, CD1575900.

If deposits are found in critical components of the fuel system, and an engine meets three or more of the following conditions, a fuel detergent additive is required to improve the dispersancy of the fuel.

- The engine is used in a hybrid power train.
- The average vehicle speed is 11 km [ 7 mi ] per hour or less.

- The engine exhaust is equipped with a diesel particulate filter.
- The fuel used is 50 percent or more diesel number 1 (D1).

For customers utilizing fuel mixtures 50 percent or more, Ultra Low Sulfur D1, the fuel oxidation stability should be above 20 hours using test standard EN 15751. For fuels falling below 20 hours, additives are required to improve the oxidation stability of the fuel.

### **⚠ CAUTION ⚠**

**Over use of fuel additives can cause adverse effects such as fuel filter plugging and reduced aftertreatment life.**

There are no known additives that increase the power or improve the efficiency of a properly maintained engine. Some fuel additives can be harmful to the engine or exhaust aftertreatment system. Fuel additives containing ash (incombustible material) will cause combustion chamber deposits. In addition, when fuel additives containing ash are used on engines equipped with a diesel particulate filter, the filter will fill up with the incombustible material, which can **not** be removed by performing a stationary regeneration. The engine may try to regenerate more frequently, but the regenerations will be unsuccessful, and the filter will continue to plug. The inlet surface of the diesel particulate filter can also be covered with an orange or rust colored substance. When these symptoms occur, the diesel particulate filter **must** be replaced.

Ash can consist of a variety of incombustible materials, but most commonly will be metals, such as iron. Cummins Inc. requires that there be no detectable ash in the fuel or fuel additives used in engines equipped with exhaust aftertreatment. Aftertreatment system failures caused by the use of additives containing ash will **not** be covered by Cummins Inc. warranty.

**Note : Cummins Inc. accepts no liability for engine damage resulting from the use of fuel additives that are not specifically approved. Consult your fuel supplier for guidance on additive use.**

#### **Water-Emulsions**

**Fuel Characteristics** - Water-emulsified diesel fuel is an alternative fuel that is made by blending water and other additives (e.g. detergents) into diesel fuel.

**Emissions** - Water-emulsified diesel fuels have been verified by EPA and some state agencies as an emissions reduction technology.

Cummins Inc. does **not** certify engines with water-emulsified fuels. Cummins Inc. does **not** warranty any emissions improvements with the use of water-emulsified fuels.

**Performance Issues** - Water emulsified fuels have lower energy content than Number 2 diesel fuel. Customers **must** expect at least a 15 percent power reduction and a 15 percent fuel consumption increase when water-emulsified fuels are used. Because of the lower energy content in water-emulsified diesel fuels, engines running on water-emulsified diesel fuels can require idle governor adjustments to prevent engine stalling.

**Durability Issues** - Many fuel system components in Cummins® engines are made of materials that are susceptible to corrosion from water in fuel. Prolonged exposure to water in fuel can result in fuel system component failures from corrosion.

Vehicle System Issues - Some water-emulsified diesel fuel suppliers recommend the removal of the fuel-water separator from the vehicle's fuel system. Removal of the fuel-water separator violates Cummins Inc. engine installation requirements.

Since water is a significant component of water-emulsified diesel fuels, conductivity sensors that detect water in fuel will **not** function properly with water-emulsified diesel fuels.

Some water-emulsified diesel fuels use a surfactant in the emulsifier. Surfactants can strip the fuel tank and fuel lines of deposits, resulting in fuel filter plugging. Fuel filters **must** be monitored closely during the initial use of water-emulsified diesel fuels.

Water-emulsified diesel fuels can **not** remain static for more than a month in storage or in vehicle fuel tanks. Most water-emulsified diesel fuel storage facilities are required to have circulation pumps for daily or weekly agitation. Engines operating on water-emulsified diesel fuel **must** be operating for at least 15 minutes every 30 days to avoid fuel-water separation in the vehicle fuel tank and in the engine fuel system.

Cummins Inc. Engine Warranty - Cummins Inc. Engine Warranty covers failures that are a result of defects in material or factory workmanship. Engine damage, service issues, and/or performance issues determined by Cummins Inc. to be caused by the use of water-emulsified diesel fuel are **not** considered to be defects in material or workmanship and are **not** covered under Cummins Inc. engine warranty.

Some water-emulsified fuel suppliers provide a comprehensive warranty for fuel system failures caused by the use of water-emulsified diesel fuel. Customers are encouraged to contact the water-emulsified diesel fuel supplier to determine the warranty provisions.

## Fuel Filters

This section explains the types of fuel filters and their uses.

Cummins® engines are supplied with the latest in fuel filtration technology from Cummins Filtration™. These systems are designed to remove water and other harmful particles from the fuel before they damage the fuel pump and other engine components.

### Throw Away Canister

The standard fuel filter is the spin-on element. These filters contain a porous, pleated, chemically treated paper element that will pass fuel freely but trap impurities and sediment. When the element is serviced, it is simply detached from the fuel filter head assembly, discarded, and replaced with a new element. The element **must** be tightened to the manufacturer's specifications.

**Note :** Do **not** pour fuel from an old fuel filter into a new filter in an effort to prime the fuel system. Use **only** clean fuel to prime the fuel system. It is **not** necessary to add fuel to a new filter if the engine is equipped with an electric fuel transfer pump. Fuel systems on these engines can be primed by turning the vehicle keyswitch ON and OFF several times to activate the fuel transfer pump.

## CAUTION

**Overtightening will distort the filter cartridge or crack the filter head. Do not use a filter element that has been dented or damaged prior to, or during, installation.**

## Replaceable Element Type

Another type of fuel filter used on Cummins® engines has a replaceable pleated paper element. This type of filter is often recommended or required for use as a first stage of filtration to provide additional water separation and/or fine particle removal.

## Fuel-Water Separators

Water can enter diesel fuel at various locations along the supply chain, and becomes a serious issue when present as free water. It contributes to corrosion, biological contamination, and fuel system malfunctions. Entry points include:

- **As free water** due to ingress as result of heavy rainfall or cracks in equipment
- **As dissolved (emulsified) water** during fuel refining or delivery (this may become free water further down the supply chain if the fuel is cooled so much that it reaches a saturation point)
- **As water vapor** (moist air) through vents followed by condensation on tank walls, including vehicle tanks.

Water in diesel fuel is normally present as both free and emulsified water. Free water settles to the fuel tank bottom, where it can be drained. Emulsified water stays in suspension where it can enter the fuel lines, fuel pump, and injectors.

Free and emulsified water can be removed from the fuel. Integral fuel filter and water separators are available that remove both free and emulsified water with varying degrees of efficiency. The standard fuel filter does remove some free and emulsified water, but with low efficiency. Due to the above facts and the importance of removing water from fuel for fuel system integrity, Cummins Inc. has increased the requirements for free water and emulsified water removal. The fuel-water separator or fuel filter and water separator combination **must** remove a minimum of 93 percent of free water (per SAE J1839) and 93 percent of emulsified water (per SAE J1488) (with incoming 2500 ppm water & 175 ppm max effluent) with the test fuel rated at IFT =  $12 \pm 1.5$  mN/m per ASTM D971. Fuel-water separator filters produced by Cummins Filtration™ meet or exceed these requirements.

Cummins Inc. recommends that a fuel-water separator be installed on all Cummins® engines, and strongly recommends using Cummins Filtration™ fuel-water separators that utilize StrataPore™ filter media. These StrataPore™ filters provide high efficiency removal of harmful particles and both free and emulsified water.

Fuel-water separators should be checked on a daily basis and drained into an appropriate disposal container when free water is noted. If water is indicated by a water-in-fuel (WIF) sensor, water **must** be drained immediately to prevent damage to the fuel system components. Water should **not** be allowed to fill the bowl.

**Note : The drained fluids (mixture of water and fuel) must be properly disposed of according to regulations.**

### CAUTION

**If the water level in the fuel water separator is allowed to reach the fuel filter element, water can be forced through the filter and cause corrosion and failure of sensitive components in the fuel system.**

## Fuel Filter Maintenance

Fuel filters **must** be changed periodically to prevent restriction of fuel flow from the fuel tank to the fuel pump. Fuel restriction will increase over time as sediment gets collected in the filter media. Sediment could possibly consist of rust, dirt, dust, oxidation products, and biological growth.

Change fuel filters as recommended by the appropriate Cummins® Engine Owner's Manual or Operation and Maintenance Manual. When operating under severe conditions, additional fuel filter changes can be required. To determine if this is necessary, fuel filter restriction **must** be checked. Refer to the appropriate Cummins® Engine Service Manual for fuel filter restriction checking procedures. After checking the restriction a few times, a maintenance schedule for fuel filter changes can be established for each type of operation.